

Lecture 7: Multiple Access Protocols

EC 441 — Introduction to Computer Networking

Boston University, Spring 2026

Learning Objectives

By the end of this lecture, you should be able to:

1. Explain the multiple access problem and why it arises on shared broadcast channels
2. Describe channel partitioning approaches (TDMA, FDMA) and their efficiency limitations
3. Describe taking-turns approaches (polling, token passing) and their tradeoffs
4. Derive the throughput expressions for pure ALOHA and slotted ALOHA and identify the maximum efficiency of each
5. Explain CSMA and how carrier sensing improves on ALOHA
6. Describe how CSMA/CD works and why a minimum frame size is required
7. Identify which MAC protocol is used in Ethernet vs. WiFi

— Session 1 (40 minutes) —

1 The Multiple Access Problem

1.1 Setup

A **broadcast channel** (also called a shared medium) is one in which every node's transmission is received by all other nodes simultaneously. Classic examples include:

- A coaxial cable segment (classic Ethernet before switches)
- A WiFi radio channel (all nodes share the same spectrum)
- A satellite uplink shared by many ground stations

When two or more nodes transmit at the same time, their signals overlap at every receiver. The result is a **collision**.

1.2 Channel Directionality: Simplex, Half-Duplex, Full-Duplex

Before examining MAC protocols in detail, it is useful to classify channels by their directionality. A **simplex** channel carries traffic in one direction only (e.g., broadcast radio or traditional cable television). A **half-duplex** channel supports both directions, but not simultaneously — only one end may transmit at a time (classic Ethernet on coaxial cable, WiFi, or a walkie-talkie radio). A **full-duplex** channel supports simultaneous transmission in both directions (telephone circuits, switched Ethernet, modern fiber links).

The multiple access problem is fundamentally a **half-duplex** problem. On a shared half-duplex medium, at most one node can transmit at any instant — a second simultaneous transmission causes a collision. On a full-duplex point-to-point link, each direction travels on a logically separate path, so collisions cannot occur; no MAC protocol is required beyond “transmit when you have data.” This distinction will reappear at the end of the lecture: classic bus/hub Ethernet is half-duplex and uses CSMA/CD, while modern switched Ethernet provides each device a dedicated full-duplex link to its switch port, making CSMA/CD unnecessary in practice.

1.3 What Happens in a Collision

In the physical layer, a collision is a superposition of multiple signals. When the desired signal power is S and the interfering signal power is I , the receiver must work with a signal-to-interference ratio S/I . For a binary modulation scheme in an AWGN channel, the bit error probability is approximately $Q(\sqrt{S/I})$, where $Q(\cdot)$ is the Q-function (the tail probability of a standard normal distribution).

For $S/I = 0$ dB (i.e., $S = I$), we get $Q(1) \approx 16\%$ bit error rate — far too high for useful communication. In practice, a receiver needs S/I well above 0 dB (often 10 dB or more with margin) to decode reliably. Therefore, a collision effectively destroys both frames.

1.4 What We Need: A MAC Protocol

A **multiple access (MAC) protocol** is a distributed algorithm that determines when each node is allowed to transmit and what to do after a collision. An ideal MAC protocol for a channel of rate R satisfies:

1. When **one node** has data to send, it gets the full channel rate R .
2. When M **nodes** have data, each gets roughly R/M on average.
3. The protocol is fully **decentralized** — no central coordinator and no shared clock (beyond what is strictly necessary).
4. The protocol is **simple** to implement.

No protocol simultaneously achieves all four ideals, but the three families described next each make different tradeoffs.

1.5 Three Families

Family	Core idea
Channel partitioning	Divide the channel into pieces; give each node a dedicated piece
Taking turns	Nodes explicitly pass access rights to one another
Random access	Nodes transmit freely; detect and recover from collisions

2 Channel Partitioning

2.1 TDMA — Time Division Multiple Access

In TDMA, time is divided into **frames**, each of which is subdivided into N fixed-length **slots**. Each node is permanently assigned one slot per frame and may only transmit during that slot.

- **Guaranteed rate:** Each node gets exactly R/N , always.
- **No collisions:** Transmissions are separated in time.
- **Efficiency problem:** If only one node has data, $N - 1$ slots per frame go empty. Channel utilization is $1/N$.
- **Used in:** GSM cellular, satellite links, some industrial networks.

2.2 FDMA — Frequency Division Multiple Access

In FDMA, the available bandwidth W is divided into N frequency bands. Each node is permanently assigned one band.

- Symmetric to TDMA: same efficiency limitations, same no-collision guarantee.
- Idle nodes waste their allocated spectrum.
- **Used in:** FM radio, cable TV, traditional analog cellular (AMPS).

2.3 CDMA — Code Division Multiple Access (brief)

CDMA assigns each node a unique **spreading code** that is mathematically orthogonal to all other nodes' codes. All nodes transmit simultaneously across the full bandwidth; receivers separate signals by correlating with the correct code. CDMA is used in 3G cellular (WCDMA) and GPS, and is not covered in depth in this course.

2.4 Summary Critique

Channel partitioning is well-suited to **heavy, symmetric, predictable** loads. Most LAN and internet traffic is **bursty** — nodes alternate between long idle periods and short bursts of activity. Under such conditions, partitioned capacity is wasted on idle nodes. This motivates the next two families.

3 Taking Turns

3.1 Polling

A **coordinator node** polls each other node in round-robin order, sending a brief message that grants permission to transmit one frame. After the polled node transmits (or declines), the coordinator polls the next node.

- **No collisions.**
- **Polling overhead:** One control message per poll, even for idle nodes.

- **Coordinator is a single point of failure.**
- **Used in:** Bluetooth (primary/secondary model), industrial fieldbus protocols where deterministic latency is required.

3.2 Token Passing

In token passing, a special frame called the **token** circulates around a logical ring connecting all nodes. A node may transmit frames only while it holds the token. After transmitting (or if it has nothing to send), it passes the token to the next node in the ring.

Node A $\rightarrow$ Node B $\rightarrow$ Node C $\rightarrow$ Node D $\rightarrow$ Node A $\rightarrow \dots$

- **No collisions.**
- **Predictable worst-case latency:** A node waits at most one full ring circuit for the token. This makes token passing attractive for real-time and industrial systems.
- **Token loss:** If a node crashes while holding the token, the ring stalls. Recovery protocols add complexity.
- **Ring failure:** A single failed node can break the logical ring.
- **Historically:** Token Ring (IEEE 802.5) and FDDI (a fiber-optic backbone technology) used this approach — switched Ethernet dominates today.

4 Random Access: Pure ALOHA

4.1 Motivation and History

The ALOHA protocols were developed at the University of Hawaii in the early 1970s for **ALOHAnet**, a packet radio network connecting the Hawaiian islands via satellite.¹ Multiple ground stations shared a single satellite uplink. Because no centralized coordinator was feasible, they needed a simple distributed protocol. The results established the theoretical foundation for all modern random-access protocols.

4.2 Pure ALOHA

Rule: If you have a frame ready to transmit, transmit it immediately. If a collision occurs (detected by the absence of an acknowledgment), wait a random time and retransmit.

This is maximally simple: no synchronization, no listening before transmitting.

4.3 Collision Window

Let T be the transmission time of one frame (the time to put all bits on the wire). A frame transmitted starting at time t occupies the channel during $[t, t + T]$.

A collision occurs if any other node begins transmitting during the window $[t - T, t + T]$, a window of duration $2T$. (A late-starting frame overlaps the tail of our frame; an early-starting frame's tail overlaps our start.)

The collision window for pure ALOHA is $2T$.

¹<https://en.wikipedia.org/wiki/ALOHAnet>

4.4 Throughput Analysis

Assume the aggregate number of transmission attempts per frame-time T follows a **Poisson** distribution with mean G (the *offered load*). A given transmission succeeds if no other attempt falls within the $2T$ collision window, i.e., no other attempt in a window of duration $2T$:

$$P(\text{success}) = e^{-2G} \quad (1)$$

The **throughput** S (average number of successful transmissions per frame-time) is:

$$S = G e^{-2G} \quad (2)$$

To find the maximum, differentiate with respect to G and set to zero:

$$\frac{dS}{dG} = e^{-2G} - 2G e^{-2G} = e^{-2G}(1 - 2G) = 0 \implies G = \frac{1}{2}$$

Pure ALOHA Maximum Throughput

$$S_{\max} = \frac{1}{2} e^{-1} = \frac{1}{2e} \approx 0.184$$

At most **18%** of the channel capacity can be used productively.

The other 82% is lost to collisions or idle time. This poor efficiency motivates both slotted ALOHA and carrier sensing.

— Session 2 (40 minutes) —

5 Slotted ALOHA

5.1 The Improvement

Key idea: Require all nodes to synchronize to a common slot clock, with slot duration equal to T (one frame-time). Transmissions may only begin at slot boundaries.

This halves the collision window: two frames can only collide if they begin in the *same slot*, so the collision window shrinks from $2T$ to T .

Rule: If you have a frame, transmit it at the start of the next slot. If a collision occurs, independently retransmit in each future slot with probability p .

5.2 Throughput Analysis

With offered load G attempts per slot:

$$P(\text{success}) = e^{-G} \implies S = G e^{-G} \quad (3)$$

Maximizing: $dS/dG = e^{-G}(1 - G) = 0 \implies G = 1$.

Slotted ALOHA Maximum Throughput

$$S_{\max} = e^{-1} = \frac{1}{e} \approx 0.368$$

Exactly **double** the pure ALOHA maximum.

5.3 What Happens at Peak Load ($G = 1$)?

Slot outcome	Probability	Explanation
Successful frame	$1/e \approx 37\%$	Exactly one attempt this slot
Empty slot	$1/e \approx 37\%$	Zero attempts
Collision	$1 - 2/e \approx 26\%$	Two or more attempts

At best, about a third of slots are productive, a third are empty, and a quarter are collisions. The slot synchronization requirement (non-trivial in a distributed system) is a practical drawback.

5.4 Comparison: ALOHA Protocols

Protocol	Max S	Collision window	Sync required
Pure ALOHA	$1/(2e) \approx 18\%$	$2T$	No
Slotted ALOHA	$1/e \approx 37\%$	T	Yes (slots)

Both ALOHA variants suffer from a fundamental problem: nodes transmit without knowing whether the channel is currently in use. On a LAN, the propagation delay τ between any two nodes is much smaller than T . Can we exploit that fact? Yes — by listening before transmitting.

6 CSMA — Carrier Sense Multiple Access

6.1 The Key Insight

Carrier sense: Before transmitting, listen to the channel. If another node is transmitting (“carrier detected”), defer.

This appears simple, but it has a profound effect on efficiency. Collisions can now only occur when two nodes both sense the channel idle within one propagation delay τ of each other. Since $\tau \ll T$ for typical LANs, the collision window is tiny.

Formally, efficiency is related to $a = \tau/T$: the smaller a is (i.e., the longer frames are relative to propagation delay), the better CSMA performs.

6.2 CSMA Variants

1-persistent CSMA If the channel is idle, transmit immediately with probability 1. If busy, wait continuously and transmit the instant the channel goes idle. *Problem:* If two nodes are both waiting for the channel to clear, they will both transmit at the first idle instant — a guaranteed collision.

Non-persistent CSMA If the channel is busy, wait a *random* time and sense again, rather than listening continuously. Less aggressive; wastes some time after the channel clears, but reduces the probability of simultaneous transmissions.

p -persistent CSMA Used in slotted systems. If the channel is idle, transmit with probability p and defer to the next slot with probability $1 - p$. Tunable tradeoff between the above extremes.

6.3 Why Collisions Still Happen

Even with carrier sensing, collisions occur when two nodes both sense the channel idle at nearly the same instant. If node A begins transmitting at time $t = 0$, its signal takes τ seconds to reach node B. If B senses the channel at any time $t < \tau$, it hears silence and may begin transmitting — starting a collision that neither node detects until A's signal reaches B and B's signal reaches A.

Figure 1 illustrates this in a space-time diagram, where each frame's propagation appears as a parallelogram. The worst case is when B begins just ε before A's signal arrives.

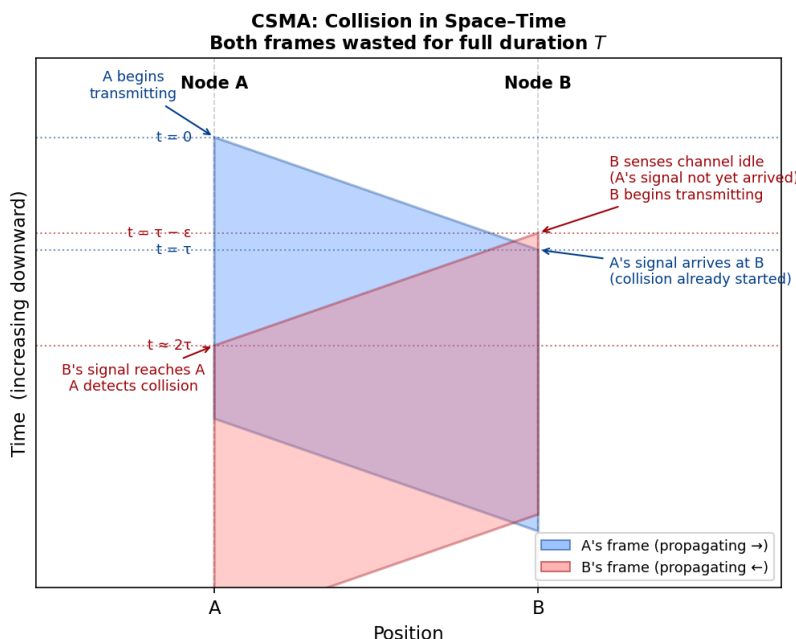


Figure 1: CSMA collision in a space-time diagram. Node A starts transmitting at $t = 0$. Node B senses the channel idle at $t = \tau - \varepsilon$ (just before A's signal arrives) and begins transmitting. Both frames are wasted for their full duration T . A detects the collision only when B's signal returns at $t \approx 2\tau$.

Once a collision begins, both frames are sent for their full duration T before either node knows. This motivates **collision detection**.

7 CSMA/CD — Collision Detection

7.1 The Enhancement

Collision detection (CD): While transmitting, also listen to the channel. If the signal on the channel differs from what the node is sending, a collision is occurring. **Abort immediately** — no need to transmit the rest of the wasted frame. This is the basis of **classic (shared) Ethernet** (IEEE 802.3).²

The key improvement: instead of wasting all of T on a collided frame, a node aborts after at most 2τ (the round-trip propagation time). Since $\tau \ll T$, this is a dramatic reduction in wasted channel time.

²https://en.wikipedia.org/wiki/Carrier-sense_multiple_access_with_collision_detection

7.2 CSMA/CD Algorithm

1. If channel idle $\Rightarrow$ transmit the frame (1-persistent).
2. If channel busy $\Rightarrow$ wait until idle, then transmit.
3. While transmitting, **monitor** the channel.
4. If a collision is detected:
 - (a) Send a **jam signal** (48 bits of a fixed pattern) to ensure every node on the segment recognizes that a collision has occurred.
 - (b) **Abort** the current frame.
 - (c) Enter **binary exponential backoff** (see below).
5. After the backoff delay, return to step 1.

7.3 Binary Exponential Backoff

After the k -th consecutive collision on a given frame, the node chooses a random wait uniformly from $\{0, 1, 2, \dots, 2^{\min(k, 10)} - 1\}$ slot times (one slot time = 2τ for Ethernet):

Collision #	Range	Max wait (slots)
1	$\{0, 1\}$	2
2	$\{0, \dots, 3\}$	4
3	$\{0, \dots, 7\}$	8
$\vdots$	$\vdots$	$\vdots$
10+	$\{0, \dots, 1023\}$	1024
16	Give up; report error to upper layer	

The exponential growth is adaptive: under light load, short backoffs keep latency low. Under heavy load, longer backoffs spread nodes out in time, reducing the rate of future collisions. This is the same intuition used in many network congestion control mechanisms.

7.4 Minimum Frame Size

Why Ethernet Has a 64-byte Minimum Frame

For collision detection to work, a node must still be transmitting when it hears the collision signal return. If the node finishes transmitting before the collision echo arrives, it has no way to know the collision occurred and will incorrectly consider the transmission successful.

Required condition:

$$T \geq 2\tau \quad \implies \quad \frac{L}{R} \geq 2\tau \quad \implies \quad \boxed{L \geq 2\tau R}$$

Classic Ethernet parameters:

- Data rate: $R = 10 \text{ Mb/s}$
- Maximum segment length: 2500 m (with repeaters)
- Propagation speed: $\approx 2 \times 10^8 \text{ m/s}$
- One-way propagation delay: $\tau = 2500 / (2 \times 10^8) = 12.5 \mu\text{s}$
- With system margins, use $\tau \approx 25 \mu\text{s}$

Minimum frame size:

$$L \geq 2 \times 25 \mu\text{s} \times 10 \text{ Mb/s} = 500 \text{ bits} = 64 \text{ bytes}$$

The **64-byte minimum Ethernet frame size** is a direct consequence of the physics of propagation on a 2500-meter network at 10 Mb/s. This constraint persists in all modern Ethernet standards even though shared segments no longer exist in practice.

7.5 CSMA/CD Efficiency

For large frames (small $a = \tau/T$), the efficiency of CSMA/CD under 1-persistent operation is approximately:

$$\eta = \frac{1}{1 + 5a} \quad (4)$$

As $a \rightarrow 0$ (frame transmission time much larger than propagation delay), $\eta \rightarrow 1$. The key takeaway: **long frames on short networks are very efficient.**

7.6 The End of Shared Ethernet

Modern Ethernet uses **switches** connected to devices by full-duplex point-to-point links. Since each link connects exactly one device to exactly one switch port, there is no shared medium — and therefore no possibility of collision. CSMA/CD is specified in IEEE 802.3 but is never exercised in practice on modern networks.

The MAC protocol for switched Ethernet is essentially: “transmit whenever you want.” The value of CSMA/CD today is historical and pedagogical, but the **frame format** and **MAC addressing** conventions it defined are still universal.

8 Preview: CSMA/CA and Wireless

8.1 Why CD Fails in Wireless

CSMA/CD requires that a transmitting node can simultaneously hear the channel and detect collisions. In wireless, this fails for two reasons:

1. Self-interference: A transmitting node's own signal is many orders of magnitude stronger than any received signal. The radio's own transmitter drowns out everything it might receive. Collision detection is physically impossible.

2. The hidden terminal problem: Even if self-interference were solved, a node may be unable to detect that a collision is occurring at the intended receiver, because the interfering transmitter is out of range.

Figure 2 shows the geometry: Node A and Node C are each in range of Node B, but not in range of each other. Both sense the channel idle (neither can hear the other) and transmit simultaneously — collision at B, undetected by both A and C.

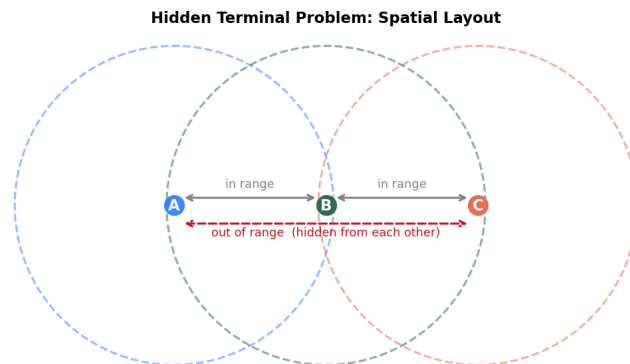


Figure 2: Hidden terminal problem: spatial layout. A and C are each within range of B but out of range of each other. Both sense the channel idle and transmit to B simultaneously.

8.2 CSMA/CA — Collision Avoidance

WiFi (IEEE 802.11) uses **collision avoidance** instead of detection. The key ideas:

- Use a random backoff **before** transmitting (not just after a collision), to reduce the probability of simultaneous transmissions.
- Optionally use an **RTS/CTS** (Request to Send / Clear to Send) handshake to “reserve” the channel before the data frame. All nodes within range of either the sender or the receiver hear the reservation and defer — solving the hidden terminal problem.
- **ACK every frame:** the absence of an ACK implies a collision or loss; the sender retransmits.

Full details of 802.11 MAC are covered in Lecture 9.

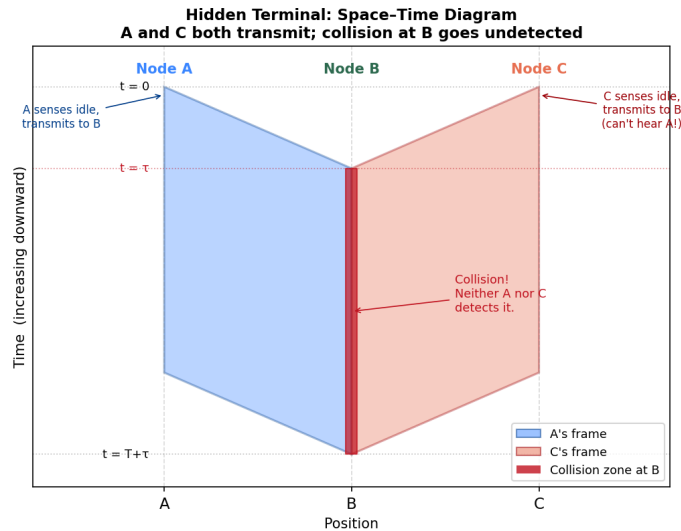


Figure 3: Hidden terminal problem: space-time diagram. Both A and C begin transmitting at $t = 0$. Their frames propagate inward and collide at B from $t = \tau$ onward. Neither A nor C can detect the collision.

Family	Protocol	Used in	Key mechanism
Partitioning	TDMA	GSM, satellite	Fixed time slots
Partitioning	FDMA	FM radio, cable	Fixed frequency bands
Taking turns	Token Ring	Historically: 802.5, FDDI	Circulating token
Random access	Pure ALOHA	Packet radio (1970s)	Transmit freely; backoff
Random access	Slotted ALOHA	Satellite, LTE random access	Slot-synchronized ALOHA
Random access	CSMA/CD	Classic Ethernet	Sense + detect + abort
Random access	CSMA/CA	WiFi (802.11)	Sense + avoid + ACK

Table 1: MAC protocol families and their principal use cases.

9 Summary

9.1 MAC Protocol Comparison

9.2 Key Numbers

Result	Value
Pure ALOHA throughput	$S = Ge^{-2G}$; $S_{\max} = 1/(2e) \approx 18\%$ at $G = 1/2$
Slotted ALOHA throughput	$S = Ge^{-G}$; $S_{\max} = 1/e \approx 37\%$ at $G = 1$
Slotted vs. pure ALOHA	Slotted is exactly $2\times$ more efficient
CSMA/CD efficiency	$\eta = 1/(1 + 5a)$ where $a = \tau/T$
Ethernet minimum frame	$L_{\min} = 2\tau R = 64$ bytes

9.3 Looking Ahead

Lecture 8 (Ethernet): Ethernet frame structure, MAC addressing, ARP, and how Ethernet switches use self-learning forwarding tables.

Lecture 9 (Wireless): 802.11 frame structure, CSMA/CA details, RTS/CTS, hidden terminal solution, and survey of WiFi generations.

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- [9] Metcalfe, R.M. and Boggs, D.R. “Ethernet: distributed packet switching for local computer networks.” *Communications of the ACM*, 19(7):395–404, 1976. (Original Ethernet paper.)

Key Formulas

Formula	Description
$S = Ge^{-2G}$	Pure ALOHA throughput
$S_{\max} = 1/(2e) \approx 18\%$	Pure ALOHA maximum (at $G = 1/2$)
$S = Ge^{-G}$	Slotted ALOHA throughput
$S_{\max} = 1/e \approx 37\%$	Slotted ALOHA maximum (at $G = 1$)
$\eta = 1/(1 + 5a)$	CSMA/CD efficiency, $a = \tau/T$
$L_{\min} \geq 2\tau R$	Ethernet minimum frame size

Interactive exploration: `lecture_07_exploration.py` (ALOHA throughput curves, CSMA/CD efficiency vs. a)